

Frontend Launch Gaps & Readiness

Gym Operations Platform — Admin Web Portal (Next.js). Verified against current code, 2026-07-18. Scope: admin web only; mobile Expo app is unbuilt (member/trainer flows §19–29 out of scope). State today: ~5k LOC, 12 admin + 5 auth pages → 99-endpoint FastAPI backend via server-side auth proxy.

1 · Launch-Blocking Missing Features

#	Feature	Spec	Status Now	Sev	Why It Blocks Launch
1	Real-time WebSocket sync	§16.4	Backend /ws exists; frontend has ZERO WS usage	Critical	Core promise ('no refresh buttons'); data stale until reload
2	MFA management UI	§5.5	Backend /auth/mfa/* exists; 0 UI refs	Critical	Enterprise tier = mandatory MFA; cannot onboard
3	Session management UI	§5.9 / Rule 8	Backend /auth/sessions + revoke; 0 UI	Critical	'Owner can revoke any session' is a security rule
4	error / loading / not-found	UX	None in app/	Critical	Any thrown fetch = white-screen crash in prod
5	Classes management page	§21, §6.4	Backend classes.py (6 ep); no page	High	Bookings, rosters, check-ins unreachable
6	Staff / Trainer page	§7,15,24	Backend staff.py (11 ep); no page	High	Can't invite trainers / set rates; payroll depends on it
7	Audit log viewer	§16.1 / R10	Backend records audits; no page	High	'Full searchable audit log' = named web capability
8	Approval queue (enrollment)	§8.3	Backend supports approved mode; no UI	High	Approved-mode gyms can't approve prospects
9	Middleware auth guard	Security	Per-layout redirect only; no middleware.ts	Medium	Fragile; no central route protection / CSP
10	Automated tests	Quality	Zero frontend tests	Medium	No regression net for money-handling product

2 · Spec Domain → Frontend Coverage

Domain	Backend	Frontend	Coverage	Gap
Auth (login/register/reset/magic)	20 ep	5 pages	~85%	No MFA enroll; verify resend UX
Organizations / Settings	9 ep	settings	~50%	No MFA, no sessions, no logo/accent branding
SaaS Billing	6 ep	billing	~70%	No invoice PDF; downgrade cap-warning UX
Plans	10 ep	plans	~90%	Strong; verify archive→replacement picker
Members	7 ep	members	~80%	CSV import ok; verify freeze/ban actions
Memberships	8 ep	(in members)	~40%	No dedicated subscription lifecycle view
Payments	2 ep	payments	~70%	Verify web-only refund UI (Rule 7)
Cash	2 ep	cash	~90%	Reconciliation likely covered
Receipts	5 ep	receipts	~75%	Queue ok; no image lightbox / audit reversal
Classes	6 ep	none	0%	Whole domain unreachable
Staff	11 ep	none	0%	Whole domain unreachable
Payroll	9 ep	payroll	~80%	Verify draft→lock→finalize→CSV flow
Analytics	2 ep	analytics	~70%	Charts ok; no CSV export (§16.1)
Real-time	ws	none	0%	See blocker #1

3 · Edge Cases & Edge Problems

Area	Edge Case / Problem	Current Risk	Fix
Auth	Access token expires (15m) mid-POST	Refresh-on-401 exists but untested for bodies	Verify body re-sent on retry; add test
Auth	Refresh token (7d) expires → refresh fails	Raw 401 may bubble	Global redirect-to-login, not crash

Area	Edge Case / Problem	Current Risk	Fix
Auth	Vague login errors (Rule 4)	Backend vague; frontend may leak field	Audit error rendering
Idempotency	Double-tap Pay / submit	Proxy forwards key; who makes UUID?	Client generates UUID + disables btn (\$13.9)
Multi-org	Org switch mid-session (\$14.3)	Switcher exists; stale WS/data?	Full reload + JWT re-scope
Payments	Stripe in STUB mode (stripe_live=false)	Real card form won't work	Wire Stripe Elements + live keys
SaaS	Read-only (d6) / suspended (d30) states	Frontend may not render them	Add banner + disable mutations
UX	Empty states (new gym, 0 rows)	Some EmptyState used	Audit every list page
Concurrency	Two admins edit same plan	No optimistic-lock UX	Stale-write warning
Locale	Org timezone / currency vs browser	May render wrong TZ/currency	Centralize in format.ts

4 · Common Cross-Cutting Problems

Problem	Present?	Impact	Action
error.tsx / global-error.tsx	Missing	Prod crash = blank page	Add root + per-segment
loading.tsx skeletons	Missing	Janky perceived perf	Add per-segment
not-found.tsx	Missing	Ugly 404	Add
CAPTCHA on signup forms	Backend only	Bot signups	Wire hCaptcha/Turnstile (\$7.3)
CSP / security headers	Missing	XSS surface	Add via next.config headers
Sentry / error monitoring	Missing	Blind in prod	Wire Sentry (spec Monitoring)
Accessibility (a11y) pass	Unknown	Legal/UX risk	Labels, focus, contrast
Responsive (tablet/mobile-web)	Unknown	Owner on-the-go usage	Test breakpoints
Env config per environment	Hardcoded default	Staging/prod misconfig	BACKEND_URL per env
Toast / global notifications	Unknown	Silent failures	Confirm global toaster

5 · Infra / Ops Blockers (backend-side)

Item	State	Blocks?	Note
Alembic migrations	None (SQLite auto)	Yes	Need Postgres migrations
docker-compose.yml	Empty stub	Yes	—
Postgres + Redis	Not provisioned	Yes	SQLite default only
Stripe live keys (Platform+Connect)	Stubbed	Yes	For real payments
Email (Resend) domain verify	Known issue	Partial	Invites silently fail
Real OCR / GPT-4o Vision	Stubbed	No*	Receipts auto-approve fake
S3 / R2 file storage	Not wired	Partial	Receipt images, PDFs
Celery + Redis broker	Needs Redis	Partial	Workers won't run

Recommended Launch Path

- **P0 — do not ship without:** WS real-time (#1), MFA UI (#2), Session UI (#3), error/loading boundaries (#4), Stripe live wiring, Postgres + Alembic, docker-compose.
- **P1 — full product:** Classes page (#5), Staff page (#6), Audit log (#7), analytics CSV export, SaaS read-only/suspended states.
- **P2 — hardening:** tests, Sentry, CSP headers, a11y, empty-state audit, approval queue.